TECHNICAL RESEARCH PROPOSAL

Where Do England's Premier League and Championship Footballers Come From?

Detecting Birthplace Hotspots Across English Local Areas Beyond Population Density, and Their Social and Spatial Correlates

Study population	Players with Premier League or Championship careers, born 1975–2005
Main analysis	English-born players, analysed across English local areas
Discipline	Spatial statistics · Sports geography · Spatial epidemiology methods
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All data sources used in this study are openly available at no cost. Results are intended for academic and non-commercial use.

Abstract

English football invests heavily in finding young talent, yet it is unclear whether the players who reach the Premier League and Championship come from across England in proportion to population, or whether certain towns and districts produce far more than their size would suggest. This proposal sets out a reproducible, fully open-data study of players born between 1975 and 2005 who played in the top two English divisions. The main analysis focuses on English-born players and tests whether their birthplaces are spatially clustered *beyond* population, locates statistically significant hotspots and coldspots across local authority districts and regular grids, and evaluates whether deprivation, urbanisation and proximity to professional clubs and academies help explain the pattern. Player records come from Wikidata; division membership by season comes from the engsoccerdata archive, extended to the present season; expected counts are derived from Office for National Statistics live births by local authority in each player's birth year. The analysis combines inhomogeneous point-pattern statistics, Empirical-Bayes-adjusted Moran's I, Getis-Ord Gi* and LISA hotspot detection, and negative-binomial and spatial regression models. A secondary, descriptive analysis maps the countries that supply foreign-born players to the two leagues. Outputs comprise a cleaned open dataset, hotspot maps, a methods-and-results report and an interactive R Shiny dashboard.

Keywords: Premier League, EFL Championship, talent identification, birthplace effect, spatial point patterns, hotspot analysis, Getis-Ord Gi*, spatial regression, deprivation, Wikidata

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1. Introduction

1.1 Background

The Premier League and the EFL Championship are the top two tiers of English football and among the most closely followed competitions in the world. Clubs in both divisions run youth academies, and since 2012 the Elite Player Performance Plan (EPPP) has graded those academies into categories that shape where young players are recruited and trained. Despite this investment, there is little systematic, population-adjusted evidence on *where in England* top-flight and second-tier players are born.

Research on the "birthplace effect" shows that, in several sports, the size and type of community in which an athlete is born is associated with the likelihood of reaching elite level (Côté et al., 2006; MacDonald et al., 2009). Most of that work relies on broad city-size categories rather than formal spatial statistics. England offers ideal conditions for a more rigorous test: its players are well documented in open databases such as Wikidata (Vrandečić & Krötzsch, 2014), and the Office for National Statistics (ONS) publishes detailed, openly licensed population, boundary and deprivation data for small areas.

1.2 Problem Statement

Maps of where English footballers are born largely reproduce maps of where people live: large cities naturally produce more players. Without an explicit population baseline it is impossible to say whether an apparent "talent hotbed" is genuinely exceptional or simply populous. Standard mapping and regression also ignore spatial dependence, which can make hotspot claims misleading. In addition, roughly two-thirds of Premier League players are born abroad, so a study that pools all players would measure clubs' international recruitment rather than domestic talent production. There is currently no open, reproducible, multi-scale analysis that tests whether English-born Premier League and Championship players are concentrated in particular areas beyond what population predicts.

2. Literature Review

2.1 Birthplace Effects in Sport

Côté, MacDonald, Baker and Abernethy (2006) showed that athletes from mid-sized communities were over-represented among professional players in several North American sports, introducing the term "birthplace effect." MacDonald et al. (2009) reported a similar place-of-birth influence in American football. These studies establish that place matters, but they use population-size categories rather than continuous spatial methods and have not examined English football at small-area level.

2.2 Spatial Methods for Clustering and Hotspots

Ripley's K-function (Ripley, 1977) is the standard tool for detecting clustering in point patterns; its inhomogeneous extension (Baddeley et al., 2000) tests clustering against a varying background intensity such as population. For area-level rates, global Moran's I measures overall spatial autocorrelation, and the Empirical-Bayes index of Assunção and Reis (1999) corrects it for unequal population sizes. Local hotspots are identified with the Getis-Ord G_i^* statistic (Getis & Ord, 1992) and Local Indicators of Spatial Association (Anselin, 1995). Many simultaneous local tests are controlled with the false discovery rate (Benjamini & Hochberg, 1995).

2.3 Spatial Regression and Scale

When model residuals are spatially autocorrelated, spatial lag and spatial error models (Anselin, 1988) or Bayesian BYM2 models for counts (Riebler et al., 2016) give valid inference. Results from aggregated data can change with the choice of boundaries — the Modifiable Areal Unit Problem (Openshaw, 1984) — so analysis at several scales is required.

2.4 Research Gap

No published study, to our knowledge, combines (i) a complete list of Premier League and Championship players born over three decades, (ii) a birth-year-matched official population baseline, (iii) small-area analysis across England, and (iv) formal point-pattern, hotspot and spatial-regression methods, together with deprivation and club-proximity covariates. This proposal addresses that gap.

3. Aim, Objectives, Research Questions and Hypotheses

3.1 Aim

To determine whether, where and why English-born Premier League and Championship footballers are concentrated in particular areas of England beyond what population predicts.

3.2 Specific Objectives

- **O1.** Build a clean, geocoded dataset of Premier League and Championship players born 1975–2005, with each player's divisions and seasons.
- **O2.** Derive birth-year-matched expected player counts from ONS live-births data for several spatial units in England.
- **O3.** Test for clustering of English-born players' birthplaces beyond population using inhomogeneous point-pattern statistics.
- **O4.** Identify and map significant hotspots and coldspots across English local areas at several scales.
- **O5.** Estimate associations between hotspot intensity and deprivation, urbanisation and proximity to professional clubs and academies.
- **O6.** Describe the countries supplying foreign-born players to the two leagues.
- **O7.** Publish an open, reproducible code base, dataset and interactive dashboard.

3.3 Research Questions

Code	Research question	Objective(s)
RQ1	Are the birthplaces of English-born Premier League and Championship players more clustered than population alone would predict?	O1–O3
RQ2	Where in England are the statistically significant hotspots and coldspots of player production?	O4
RQ3	Is hotspot intensity associated with deprivation, urbanisation or proximity to professional clubs and academies, after controlling for population?	O5
RQ4 (desc.)	Which countries and regions supply foreign-born players to the two leagues, and how has this changed across birth cohorts?	O6

Table 1. Research questions mapped to objectives.

3.4 Hypotheses

Hypothesis	Statement	Test
H0 (null)	English-born players' births are distributed in proportion to population; every area produces players at its expected rate.	Inhomogeneous K/L envelopes; EB Moran's I
H1	Some areas of England produce significantly more (hotspots) or fewer (coldspots) players than population predicts.	Gi*, LISA with FDR correction
H2	Hotspot intensity is associated with deprivation, urbanisation and proximity to professional clubs and academies.	NB / spatial regression coefficients

Table 2. Hypotheses and statistical tests. RQ4 is descriptive and has no formal hypothesis.

4. Significance of the Study

- **Scientific:** provides the first population-adjusted, multi-scale test of the birthplace effect for English top-two-division football, using methods from spatial epidemiology.
- **Practical:** shows clubs, academies, the Premier League, the EFL and the Football Association which areas over- and under-produce players relative to population, informing scouting coverage and grassroots investment.
- **Social:** tests whether deprivation helps or hinders the route to professional football, a question of interest to local authorities and community sport.
- **Methodological:** demonstrates a reusable open-data pipeline (Wikidata + ONS + spatial statistics) applicable to other sports and countries.
- **Open science:** all data, code and results are released for reuse and verification.

5. Scope and Delimitations

Dimension	Included	Excluded
Players	Men who played for a club during a season in which it was in the Premier League or Championship	Players whose highest level was League One or below; women's football (possible future study)
Birth cohort	Born 1 Jan 1975 – 31 Dec 2005	Earlier births (outside the Premier League era); later births (careers incomplete)
Seasons	1992/93 to 2025/26 for the Premier League; matching seasons for the second tier (First Division to 2004, Championship from 2004/05)	Seasons before 1992/93
Main analysis	English-born players, analysed across England	Scotland, Wales and Northern Ireland (different statistics and deprivation indices)
Secondary analysis	Foreign-born and other UK-born players, by country of birth (descriptive)	Hotspot inference outside England
Variables	Birthplace, birth year, position, divisions and seasons, population, deprivation, urbanisation, distance to clubs and academies	Match physical metrics, wages, transfer fees, ethnicity

Table 3. Scope of the study.

Operational definitions. A *study player* is a Wikidata item with occupation "association football player" (P106 = Q937857), sex male (P21 = Q6581097), and a club membership (P54) whose start and end dates overlap a season in which that club was in the Premier League or the second tier, as recorded in the engsoccerdata archive. Where Wikidata records matches played for a membership (P1350), a value of at least one is required. The *Premier League subset* comprises players with at least one such membership in a Premier League season. *English-born* means the birthplace coordinates fall inside England.

6. Methodology

6.1 Research Design

The study is a quantitative, observational, cross-sectional spatial analysis. It follows the logic of disease-mapping studies in spatial epidemiology: observed counts of an outcome (players born in an area) are compared with counts expected from the population of that area, and departures from expectation are tested for spatial structure and explained with covariates.

6.2 Data Sources

Dataset	Provider	Resolution / period	Use	Licence
Player records (birthplace, coordinates, birth date, clubs and dates, position)	Wikidata SPARQL endpoint	Point; births 1975–2005	Main outcome data	CC0
Club division by season	engsoccerdata (R package, Curley); later seasons from published league tables and Wikidata season items	Season; package to 2015/16, compiled 2016/17–2025/26	Which clubs were in each tier each season	Open source / CC0
Live births by area of usual residence (male births where available)	ONS (via Nomis and ONS birth statistics)	Local authority; annual	Expected counts (primary baseline)	Open Government Licence v3.0
Population estimates and censuses	ONS (via Nomis)	Local authority; 1981 onward; Censuses 1981, 1991, 2001	Fallback baseline; sensitivity analysis	Open Government Licence v3.0
Boundaries: local authority districts, MSOAs, LSOAs	ONS Open Geography Portal	Vector	Aggregation units	Open Government Licence v3.0
English Indices of Deprivation	MHCLG	LSOA	Deprivation covariate	Open Government Licence v3.0
Rural-Urban Classification	ONS	LSOA / local authority	Urbanisation covariate	Open Government Licence v3.0
Population grid	GHS-POP (EU JRC)	1 km; 1975–2005	Intensity surface for point-pattern analysis	CC BY 4.0
Club grounds and academies	Wikidata (home venue P115 → P625); published EPPP category lists (2012 onward)	Point; by season	Distance covariates	CC0 / public information
Country populations (secondary analysis)	UN World Population Prospects	Country; annual	Players per million by country	CC BY 3.0 IGO

Table 4. Data sources. All are free to access.

6.3 Baseline: Births by Area and Year

The baseline for expected counts is the number of **live births** in each area in each birth year — male births where ONS publishes them by sex — rather than total population. Births measure the pool of children from which each cohort of players came; total population is distorted in areas with many students or retirees, which would otherwise appear as false coldspots.

Where local-authority births are not available on current boundaries for a given year (most likely for the earliest birth years), the baseline falls back to ONS mid-year population estimates, using 1981 estimates for players born 1975–1980, apportioned to

current boundaries using the 1 km GHS-POP grid where necessary. Every fallback is recorded, and a sensitivity analysis repeats the main results with the total-population baseline.

6.4 Outcome Measure

For each spatial unit *i*, the expected count is the sum over birth years *y* of the unit's births multiplied by England's player rate per birth in that year:

$$E_i = \Sigma_y B_{i,y} \times (N_y / B_{England,y})$$

where $B_{i,y}$ is the number of births in unit *i* in year *y* and N_y is the number of English-born study players born in year *y*. The **Standardised Ratio** is $SR_i = O_i / E_i$, where O_i is the observed count. SR = 1 indicates production equal to expectation; SR = 2 indicates twice the expected number.

6.5 Spatial Units

Unit	Approximate number in England	Role
Local authority districts (current boundaries)	≈ 300	Main unit for hotspot analysis and regression
Middle-layer super output areas (MSOAs)	≈ 6,800	Optional extension: fine-scale sensitivity analysis, only if time allows
Regular hexagon grid, 10 km and 20 km	≈ 1,500 and ≈ 400	Boundary-free robustness check
Regions (ITL1)	9	Summary reporting

Table 5. Spatial units of analysis.

6.6 Data Processing Pipeline

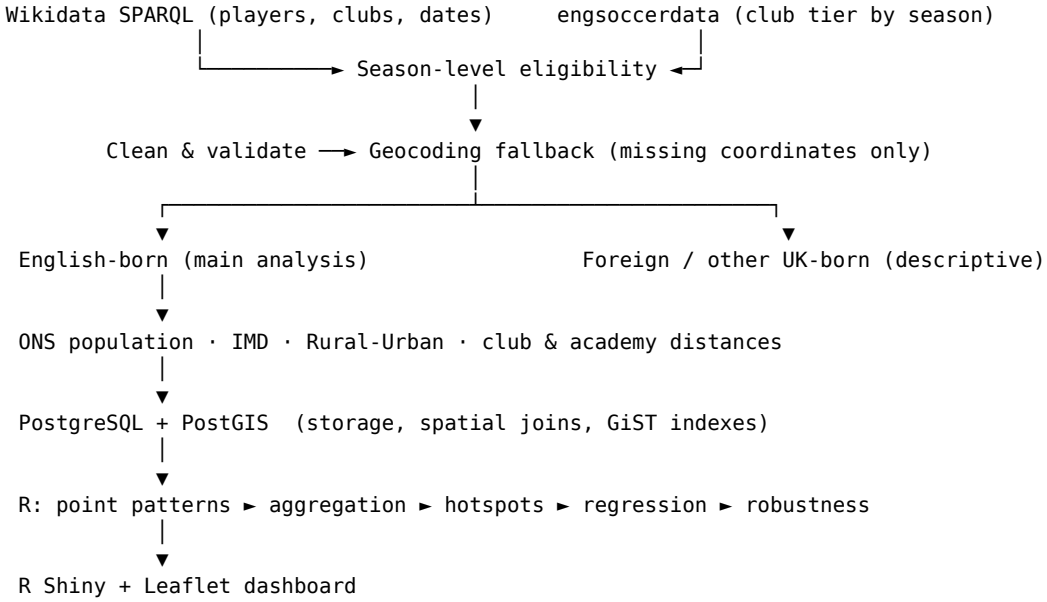


Figure 1. Data processing and analysis pipeline.

Cleaning rules.

- Retain records with birth dates between 1975-01-01 and 2005-12-31.

- Retain players with at least one club membership overlapping a Premier League or second-tier season.
- Deduplicate on Wikidata QID.
- Discard coordinates falling in the sea or outside the recorded birth country.
- Exclude records resolved only to country level from the spatial analyses (retained for the descriptive analysis).
- Write every excluded record and its reason to an exclusion log.

Season assignment rule. Wikidata club dates are often recorded to the year only. To avoid case-by-case judgements, the following rule is fixed before any data are inspected:

- A year-only **start** date Y is assigned to the season beginning in August of Y (e.g., 2004 → 2004/05).
- A year-only **end** date Y is assigned to the season ending in May of Y (e.g., 2004 → 2003/04).
- A membership that starts and ends in the same year is assigned to the season ending in that year if the start is year-only, otherwise to the season containing the full date.
- Full dates are assigned to the season that contains them; dates in June or July belong to the following season.
- Any membership whose eligibility depends on a year-only date in a season when the club changed division (promotion or relegation) is flagged as *ambiguous*. Main results include ambiguous cases; a sensitivity run excludes them.

Validation of league eligibility. A random sample of 200 players is checked manually against published season squad lists to estimate the false-inclusion and false-exclusion rates of the Wikidata-based definition, including the error rate of the season assignment rule for ambiguous cases.

Geocoding fallback. Only for birthplaces lacking coordinates, a Python routine (geopy / Nominatim, rate-limited to one request per second) queries town, then county, then country, and records the resolution level achieved.

6.7 Coordinate Reference Systems

Operation	CRS / method	Rationale
Storage	EPSG:4326 (WGS 84)	Universal interchange standard
Distances, densities, grids, areas	EPSG:27700 (British National Grid, metres)	Official projection for Great Britain; accurate distances and areas
Web display	EPSG:3857 (Web Mercator)	Display only — never used for measurement

Table 6. Coordinate reference systems.

All geometry columns carry GiST indexes. Because many players share an identical town-centre coordinate, points are jittered by up to ±500 m for point-pattern analysis only; aggregated analyses use original coordinates.

6.8 Analytical Methods

(a) Exploratory analysis. Players per 100,000 population by region and local authority; kernel density surfaces of player births and of population; and a relative-risk surface (player density ÷ population density) estimated with *spatstat::relrisk*.

(b) RQ1 — clustering beyond population. The inhomogeneous K and L functions (*spatstat::Kinhom*) are estimated within the England boundary, using the population surface as the background intensity. Global envelopes from 999 Monte Carlo simulations of births distributed in proportion to population form the null model; an observed curve above the envelope indicates clustering beyond population.

(c) RQ2 — hotspot location. Standardised Ratios are Empirical-Bayes smoothed to stabilise estimates in small-population units. Global autocorrelation is measured with the EB Moran's I (*spdep::EBImoran.mc*, 999 permutations). Local hotspots and coldspots are identified with Getis-Ord G_i^* (*spdep::localG_perm*) and Local Moran's I (LISA), classifying units as High-High, Low-Low, High-Low or Low-High. P-values are adjusted with the Benjamini-Hochberg false discovery rate (q = 0.05).

Spatial unit	Neighbour definition	Robustness alternative
Local authorities / MSOAs	Queen contiguity (row-standardised); islands such as the Isle of Wight linked to nearest neighbour	Inverse distance, 30 km cut-off
10 km / 20 km hexagons	k-nearest neighbours, k = 6	k = 4 and k = 8

Table 7. Spatial weights specifications.

(d) RQ3 — explanatory modelling. A negative binomial model is fitted with the expected count as an offset, so that coefficients describe departures from births-based expectation:

$$O_i \sim NegBin(\mu_i)$$

$$log\ \mu_i = log\ E_i + \beta_1 IMD_i + \beta_2 Urban_i + \beta_3 DistClub_i + \gamma_r$$

where IMD_i is the population-weighted deprivation score, $Urban_i$ the urban share, $DistClub_i$ the cohort-weighted distance to the nearest professional (Football League or Premier League) club *at the time each cohort was aged 12*, and γ_r region fixed effects. Every professional club ran youth development throughout the study period, so this measure applies to all cohorts.

Academy-era sub-model. Category 1 academies exist only since the Elite Player Performance Plan began in 2012, so distance to the nearest Category 1 academy ($DistAcademy_i$) is *not* used for the whole cohort. It is added in a separate model restricted to players born 1996–2005, who were aged 16 or under when EPPP began and could therefore have been developed within it. Comparing DistClub in the two cohorts shows whether proximity mattered differently before and after EPPP.

Residuals are tested for spatial autocorrelation with Moran's I. If significant, robust Lagrange Multiplier tests (*spdep::lm.RStests*) guide the choice between a Spatial Error Model (*spatialreg::errorsarlm*) and a Spatial Lag Model (*spatialreg::lagsarlm*) fitted to log EB-smoothed SR. As an optional extension, only if time allows, a BYM2 Poisson model (R-INLA) is fitted as a count-based alternative. Results are reported as rate ratios with 95% confidence (or credible) intervals, the spatial parameters ρ or λ, and AIC /

log-likelihood model comparison.

(e) RQ4 — foreign-born supply (descriptive). Counts of foreign-born and other UK-born players by country of birth, players per million national population, and changes between the 1975–1989 and 1990–2005 birth cohorts, shown as choropleth and flow maps. No hotspot inference is made, because these players are a selected subset of each country's footballers.

6.9 Robustness and Sensitivity Analyses

Check	Purpose
Multiple scales: local authority, 10 km and 20 km hexagons (MSOA optional)	Assess sensitivity to the Modifiable Areal Unit Problem
Cohort split: 1975–1989 vs 1990–2005	Test stability of hotspots across generations and before / after EPPP
Premier League subset vs all study players	Test whether hotspots differ by level reached
Truncate cohort at 2002	Check bias from incompletely recorded youngest players
Exclude the largest cities (London, Birmingham, Manchester, Liverpool, Leeds)	Check that hotspots are not simply big-city effects
Alternative neighbour definitions (Table 7)	Test sensitivity to spatial weights
Total-population baseline instead of births	Check the effect of the births baseline
Exclude ambiguous season assignments	Check the effect of year-only Wikidata dates

Table 8. Robustness checks.

6.10 Validation and Quality Assurance

- **Coordinate integrity:** unit tests confirm that round-trip transformations between EPSG:4326 and EPSG:27700 return points within 1 m.
- **Data checks:** all English-born points fall inside England; local authority births sum to the published England total within 1%; all birth years lie in 1975–2005.
- **League eligibility check:** the 200-player manual validation sample (Section 6.6).
- **Null-model test:** players simulated purely in proportion to population must yield no significant hotspots after FDR correction, showing that the pipeline does not generate false positives.
- **Code review** of the analysis scripts before results are finalised.

7. Technical Architecture and Tools

Component	Technology	Function
Development environment	GitHub Codespaces (dev container)	Reproducible cloud workspace with GDAL, GEOS and PROJ
Analysis	R: sf, terra, spatstat, spdep, spatialreg, MASS, engsoccerdata, tidyverse (INLA optional)	Spatial processing, statistics and modelling
Data acquisition	Python: requests, SPARQLWrapper, geopy, pandas	Wikidata extraction and geocoding fallback
Database	PostgreSQL 16 + PostGIS 3 (Docker service)	Spatial storage, joins and indexing
Visualisation	R Shiny + Leaflet	Interactive dashboard
Version control	Git / GitHub	Code history and collaboration
Deployment	shinyapps.io (free tier)	Public dashboard hosting

Table 9. Technology stack (all open source or free tier).

Dashboard design. (1) England hotspot map with Gi* classes, switchable between local authority, MSOA and hexagon scales, with player popups; (2) density explorer comparing player density, population density and relative risk; (3) drivers panel showing model coefficients and SR-versus-deprivation and distance plots; (4) world map of foreign-born supply; (5) filters for league (Premier League / Championship / both), birth cohort, position and region.

8. Data Management and Reproducibility

- **Versioned raw data:** each Wikidata extraction is stored as a dated Parquet snapshot so results can be reproduced after Wikidata changes.
- **Environment locking:** R packages pinned with renv; Python packages pinned in requirements.txt; database defined in docker-compose.
- **Determinism:** fixed random seeds for all permutation tests, simulations and jittering.
- **Directory structure:** data/raw, data/processed, R/, python/, sql/, app/, docs/, tests/.
- **Documentation:** data dictionary (Appendix A), exclusion log, eligibility validation report, and a README describing how to rerun the full pipeline.
- **Sharing:** code under an MIT licence; derived aggregated data under CC BY 4.0 with attribution to all upstream sources, including the Open Government Licence statement for ONS and MHCLG data.

9. Ethical and Legal Considerations

- **Nature of data:** only publicly available information about public figures (name, birth date, birthplace, clubs) is used. No sensitive personal data — such as ethnicity, health or religion — is collected.
- **Data protection:** processing follows the data-minimisation and purpose-limitation principles of the UK GDPR and Data Protection Act 2018. The research purpose is non-commercial and statistical.

- **Publication of results:** findings are reported for aggregated spatial units. The dashboard shows only fields already public on Wikidata.
- **Licensing compliance:** all upstream licences are respected and credited.
- **Interpretation:** results describe area-level patterns and must not be used to make claims about individuals or to stereotype towns or communities, particularly when discussing deprivation.

10. Limitations and Risk Management

Risk / limitation	Likelihood	Impact	Mitigation
Wikidata club dates incomplete or year-only	High	High	Pre-defined season assignment rule; ambiguity flag; 200-player validation; sensitivity run
Club membership does not guarantee an appearance	Medium	Medium	Use matches-played qualifier where present; validation sample
engsoccerdata ends at 2015/16 (≈10 seasons missing)	Certain	Medium	Compile 2016/17–2025/26 club-tier table in Week 1 from league tables and Wikidata; cross-check
Local-authority births unavailable for earliest years	Medium	Medium	Fallback to population estimates; record fallbacks; sensitivity analysis
Academy variable anachronistic for pre-EPPP cohorts	Certain	Medium	Academy distance used only for players born 1996–2005
Timeline overrun (≈135 hours)	Medium	High	BYM2 and MSOA analysis designated optional extensions
Birthplace recorded as hospital town, not home town	High	Medium	Local authority and coarser scales; stated as limitation
Birthplace differs from where a player grew up	Medium	Medium	Coarser-scale analyses; stated as limitation
Deprivation measured today, not at birth	High	Medium	Sensitivity analysis with older deprivation releases; stated as limitation
Youngest cohort (2003–05) incompletely recorded	Medium	Medium	Cohort split; sensitivity run truncated at 2002
Unstable rates in small-population areas	High	Medium	Empirical Bayes smoothing; minimum-births thresholds
Spatial autocorrelation in residuals	High	High	Spatial error / lag models; BYM2 if time allows
Modifiable Areal Unit Problem	High	Medium	Multi-scale analysis (Table 8)
Ecological fallacy	Medium	Medium	Area-level interpretation only

Table 10. Risk register.

11. Work Plan and Timeline

Activity	W1	W2	W3	W4	W5	W6
Environment set-up; club-tier table to 2025/26; Wikidata extraction; season rule; data-quality report						
ONS births and population, boundaries, IMD, urbanisation; club and academy locations; aggregation						
Exploratory analysis; KDE; inhomogeneous K (RQ1)						
EB Moran's I; Gi*; LISA hotspots (RQ2)						
Regression models; robustness checks (RQ3); foreign-born summary (RQ4)						
Dashboard; documentation; final report; deployment						
Validation and testing (continuous)						

Figure 2. Gantt chart of the six-week work plan.

Milestone	Week	Deliverable
M1	1	Clean player dataset with league-season eligibility and data-quality report
M2	2	Enriched, aggregated analysis-ready database
M3	4	Clustering and hotspot results (RQ1, RQ2)
M4	5	Model results, robustness report and foreign-born summary (RQ3, RQ4)
M5	6	Final report, public dashboard and code release

Table 11. Milestones and deliverables.

12. Budget and Resources

Item	Source	Cost
Player data	Wikidata	Free
League history	engsoccerdata	Free
Births, population, boundaries, deprivation, urbanisation	ONS; MHCLG (Open Government Licence)	Free
Population grid and country populations	EU JRC GHSL; UN World Population Prospects	Free
Software	R, Python, PostgreSQL/PostGIS, Shiny (open source)	Free
Compute	GitHub Codespaces personal free allowance	Free
Hosting	shinyapps.io free tier	Free
Total		**0**

Table 12. Budget. The project can be completed at no direct cost.

Human resources: one researcher, approximately 20–25 hours per week for six weeks (about 135 hours). **Scope control:** the core analysis uses local authorities and hexagon grids with spatial error / lag models; the MSOA-level analysis and the BYM2 (R-INLA) model are optional extensions undertaken only if the core is complete. **Contingency:** if free compute is insufficient for the 999-simulation envelopes, exploratory runs use 199

simulations and the full 999 are run only for final results.

13. Expected Outcomes and Dissemination

- A cleaned, documented open dataset of Premier League and Championship players born 1975–2005, with birthplaces, divisions and seasons.
- Formal evidence on whether English-born players' birthplaces cluster beyond population (RQ1).
- Multi-scale maps of statistically significant talent hotspots and coldspots across England (RQ2).
- Estimates of how deprivation, urbanisation and proximity to clubs and academies relate to player production (RQ3).
- A descriptive map of the countries supplying foreign-born players to the two leagues (RQ4).
- A public, interactive R Shiny dashboard and an open GitHub repository.
- A methods-and-results report suitable for submission to a sports-science or geography journal, and a portfolio case study.

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Appendix A. Data Dictionary

Variable	Type	Source	Description
player_qid	text	Wikidata	Unique Wikidata identifier (e.g., Q12345)
birth_date	date	Wikidata P569	Date of birth (1975-01-01 to 2005-12-31)
birth_place_qid	text	Wikidata P19	Identifier of place of birth
geom	geometry(Point, 4326)	Wikidata P625 / geocoder	Birthplace coordinates
geo_level	text	Pipeline	Resolution: town / county / country
birth_country	text	Spatial join	Country of birth; England flag for main analysis
position	text	Wikidata P413	Main playing position
club_qid, start, end	text, date, date	Wikidata P54 + qualifiers	Club membership periods
season, tier	text, integer	engsoccerdata + compiled 2016/17–2025/26	Club's division in each season
season_ambiguous	boolean	Derived	Eligibility depends on a year-only date in a promotion / relegation season
births_iy, baseline_type	integer, text	ONS	Births in area i, year y; 'births' or 'population' fallback
max_tier	integer	Derived	1 = Premier League, 2 = second tier
lad_code, msoa_code	text	ONS spatial join	Local authority and MSOA of birth
imd_score	numeric	MHCLG	Population-weighted deprivation score
urban_share	numeric	ONS Rural-Urban	Share of population in urban areas
dist_club_km, dist_academy_km	numeric	Derived (EPSG:27700)	Distance to nearest professional club (all cohorts) and Category 1 academy (born 1996–2005 only)
O_i, E_i, SR_i	numeric	Derived (per unit)	Observed, expected and standardised ratio

Table A1. Core variables.

Appendix B. Example Wikidata Query

```
SELECT ?player ?birthDate ?place ?coord ?club ?start ?end ?apps WHERE {
  ?player wdt:P106 wd:Q937857 ;         # association football player
          wdt:P21  wd:Q6581097 ;        # male
          wdt:P569 ?birthDate ;
          wdt:P19  ?place ;
          p:P54    ?m .                 # club membership statement
  ?m ps:P54 ?club .
  OPTIONAL { ?m pq:P580  ?start . }     # start time
  OPTIONAL { ?m pq:P582  ?end . }       # end time
  OPTIONAL { ?m pq:P1350 ?apps . }      # matches played
  VALUES ?club { wd:Q9617 wd:Q50602 }   # batch of clubs, e.g. Arsenal, Man City
  OPTIONAL { ?place wdt:P625 ?coord . }
  FILTER (YEAR(?birthDate) >= 1975 && YEAR(?birthDate) <= 2005)
}
```

Queries are run in batches of clubs (every club that has been in the top two divisions since 1992/93, listed from engsoccerdata) to stay within the endpoint's 60-second limit. Membership periods are then matched to seasons in which each club was in tier 1 or 2. Results are merged and deduplicated on player QID.

Appendix C. Glossary

Term	Meaning
Standardised Ratio (SR)	Observed players ÷ players expected from births in the area
EPPP	Elite Player Performance Plan: the Premier League's academy system, introduced in 2012
Local authority district	The main tier of local government areas used by ONS
MSOA	Middle-layer Super Output Area: ONS statistical area of about 5,000–15,000 residents
IMD	Index of Multiple Deprivation: the official measure of relative deprivation in England
Inhomogeneous K/L function	Measures clustering of points relative to a varying background intensity
Empirical Bayes smoothing	Shrinks unstable rates in small populations towards an overall mean
Moran's I	Global measure of whether similar values are located near each other
Getis-Ord G_i^*	Local statistic identifying significant clusters of high (hot) or low (cold) values
LISA	Local Indicators of Spatial Association; classifies units as High-High, Low-Low or outliers
Spatial lag / error model	Regression models that account for spatial dependence in outcomes or residuals
BYM2	Bayesian spatial model for area counts with structured and unstructured random effects
MAUP	Modifiable Areal Unit Problem: results changing with the choice of spatial units
FDR	False Discovery Rate correction for many simultaneous tests

Table C1. Glossary of technical terms.